Hoang Trieu Vy LE

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EDUCATION

2020-Now **PhD Candidate** , ENS LYON, France.
Multiscale estimation and interface detection
- [Python, C++] Study the Mumford-Shah function for Multiscale estimation and interface detection (Non-convex optimization, proximal algorithms).
- [Pytorch] Study unfolded algorithms for image restoration task.

2017–2020 **Engineering student in applied mathematics**, INSA TOULOUSE, France.
Acquired :
- Signal & Image Processing
- Mathematical optimization
- Numerical modelling (Finite Elements, Finite Volume, Finite Differences)
- Partial differential equation
- Wave and propagation in random medium
- Materials and fundamental of mechanics of structures
- Fluid mechanics
- Parallel computing (HPC, OpenMPI)
- Descriptive, Inference Statistics
- General Linear Model and Generalized Linear Model
- High dimensional and Deep Learning
- Neural network with constraints

2015–2017 **Classe préparatoire**, ÉCOLE NORMALE SUPÉRIEURE , Hué, Vietnam.
Two-year intensively studied in mathematics and physics to prepare the entrance to INSA Toulouse

Research area

- Optimization, proximal algorithm, non-smooth optimization, deep unfolded algorithms
- Image analysis : image restoration, image reconstruction
- Interface detection

Publications

- Proximal Based Strategies for Solving Discrete Mumford-Shah With Ambrosio-Tortorelli Penalization on Edges. Accepted at IEEE Signal Processing Letters journal
  [Paper] [Toolbox]
- The faster proximal algorithm, the better unfolded deep learning architecture ? The study case of image denoising. Accepted at EUSIPCO 2022 Belgrade/Serbia, 29 Aug-2 Sep
  [Paper] [Toolbox]
- Algorithmes proximaux rapides déroulés pour l'analyse d'images fractales homogènes par morceaux (Fast unfolded proximal algorithms for piecewise homogeneous fractal images analysis). Accepted at GretSi Nancy 2022
  [Paper] [Toolbox] On going

EXPERIENCES

2020 **5th year Internship** , OGoxe-Méteo France Toulouse, France.

Calculation of river flow velocity based on Particle Image Velocimetry method

- [Python, Opencv] Model and create a module for river-flow velocity estimation.

Forest fire prediction

- [Python, Pytorch] Model and create a module for forest fire prediction.

2019 **5th year Project** , INSA Toulouse, France.

Neural network with constrained binary weights

- Model and study the efficiency of a Binary Neural Network.
- [Toolbox]

2019 **5th year Project** , INSA Toulouse, France.

Anomalies Detection in aerial time-series data

- Using classical method of anomalies detection : one-class SVM, GAN.

Jul-Oct 2019 **4th year Internship** , IMFT - Toulouse Fluid Mechanics Institute, France.

Parallelization in linearly transformed particles method based on Bucketing-based algorithm

- Modelling and develop the Bucketing-based algorithm for particles method
- Parallelized the mesh-less model using MPI-Fortran, Python
- Create a state-of-the-art of new model and made test-cases on supercomputer [Toolbox]

2019 **4th year Project** , INSA Toulouse, France.

Noise reduction and retrieval by modified lidar inversion method combines joint retrieval method and machine learning

- Implemented some algorithms to resolve Lidars inverse problem
- Combined Lidars retrieval method with some denoising algorithm(Kalman-Filter, Gaussian Process)

January 2019 **Extracurricular project** , INSA Toulouse, France.

Detected and transformed body movement and physical contact to sound on Processing

- Created a cooking board which could transform physical contact into sound
- Detecte and transform body movement into sound

2018 **3rd year mathematical modeling project**, INSA Toulouse, France.

Generation of random variables on python

- Implemented some algorithms to simulate different random distribution on computer

COMPUTER SKILLS

Programming Languages	python(advanced), Fortran(advanced), C/C++ (intermidiate), MatLab(intermidiate), R(intermidiate)
Programming tools	Tensorflow, Pytorch, OpenMPI, Scikit-Learn, Freefem++, OpenCV, Anaconda
Office tools	LaTeX, OpenOffice
Operating system	Linux, Microsoft Windows

HONORS

2015 **The National Competition**, *Third prize in physics*, Vietnam.

2015 **Obtaining the Odon Vallet scholarship**.

LANGUAGES

Vietnammese	Mothertongue	
French	Advanced	*TFI B2 & DELF B1 certificate*

INTERESTS

Football, Guitare

Manim, Quantum mechanics